Phone System Testing

AND OTHER FUN TRICKS

Snide / Owen @LinuxBlog
github.com/PhreakMe (Latest Slides & Code)

Mandatory Disclaimer

- The opinions expressed in this presentation and on the following slides are solely those of the presenter. There is no guarantee on the accuracy or reliability of the information provided herein.

This is for educational purposes only

Introduction

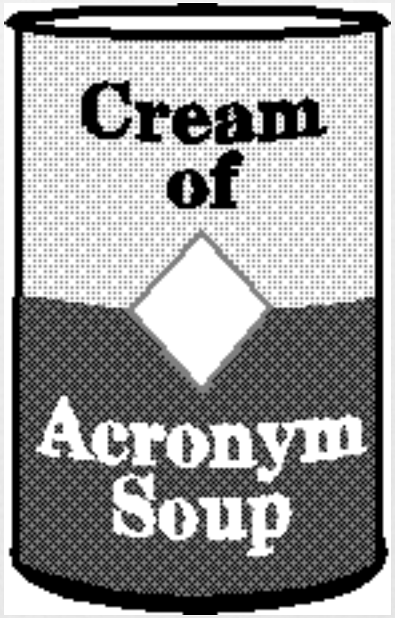
- History / Evolution
- Anatomy
- How to test
- Issue Types
- Fun Stuff

About Me

Presentations

Fun

Moved to US in 2000.



How many of you remember your childhood home phone number?

So who uses Phones?

What industries?

Particularly interesting are:

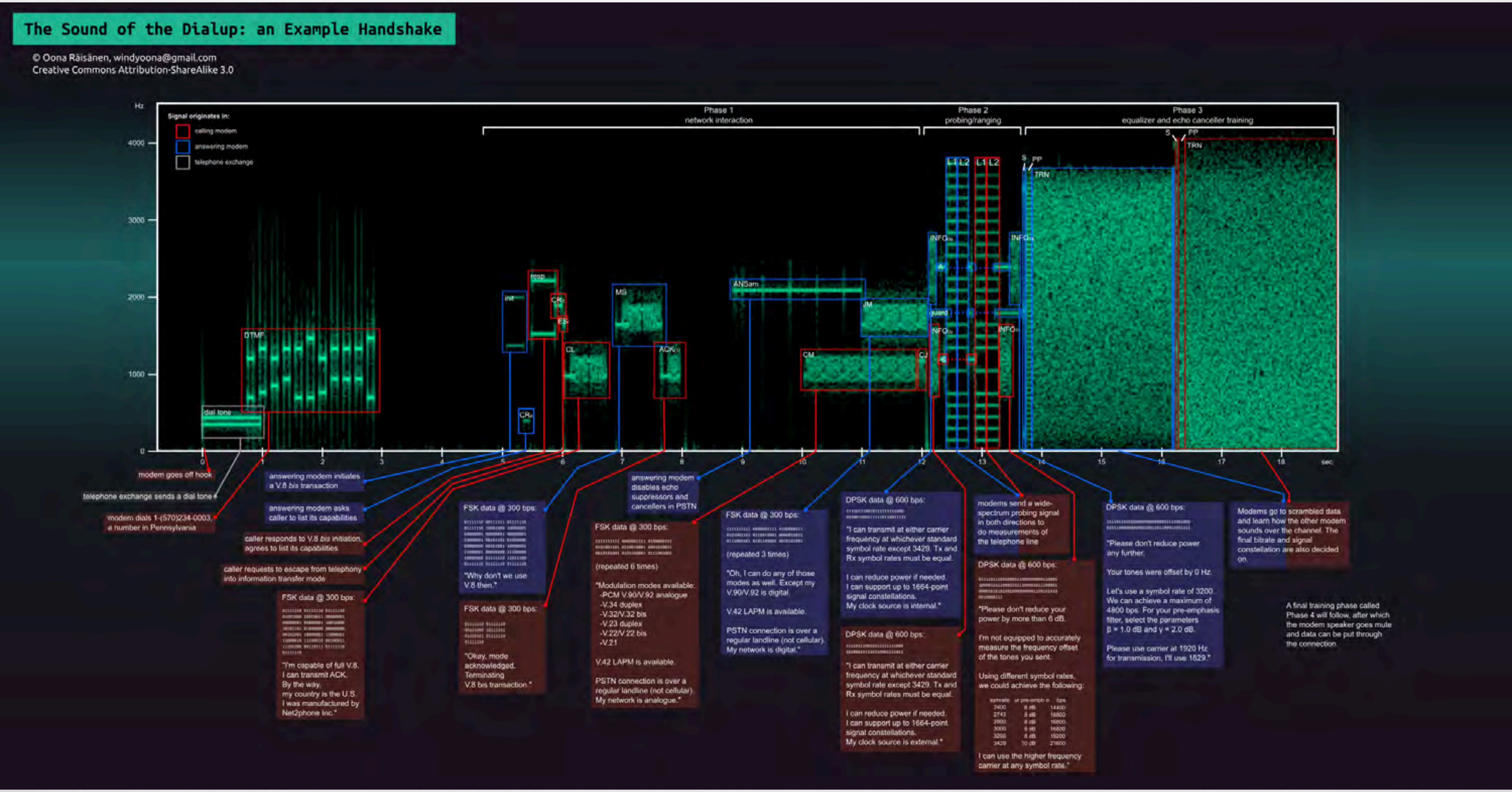
- banking/finance
- Healthcare
- Insurance
- Utilities
- Government
- Military.

History

Sorry, Wrong number DC23

Exploding The Phone (Book) 2013.

History



https://en.wikipedia.org/wiki/Dial-up_Internet_access

History

1996 ICQ, NetMeeting, SMS (UK)
1997 AIM
1998 Yahoo Messenger
1999 MSN Messenger & Asterisk
2001 TeamSpeak & MMS
2002 Yahoo Messenger Chat
2003 Skype Released - MySpace
2004 Facebook
2005 YouTube
2007 iPhone

Recent History

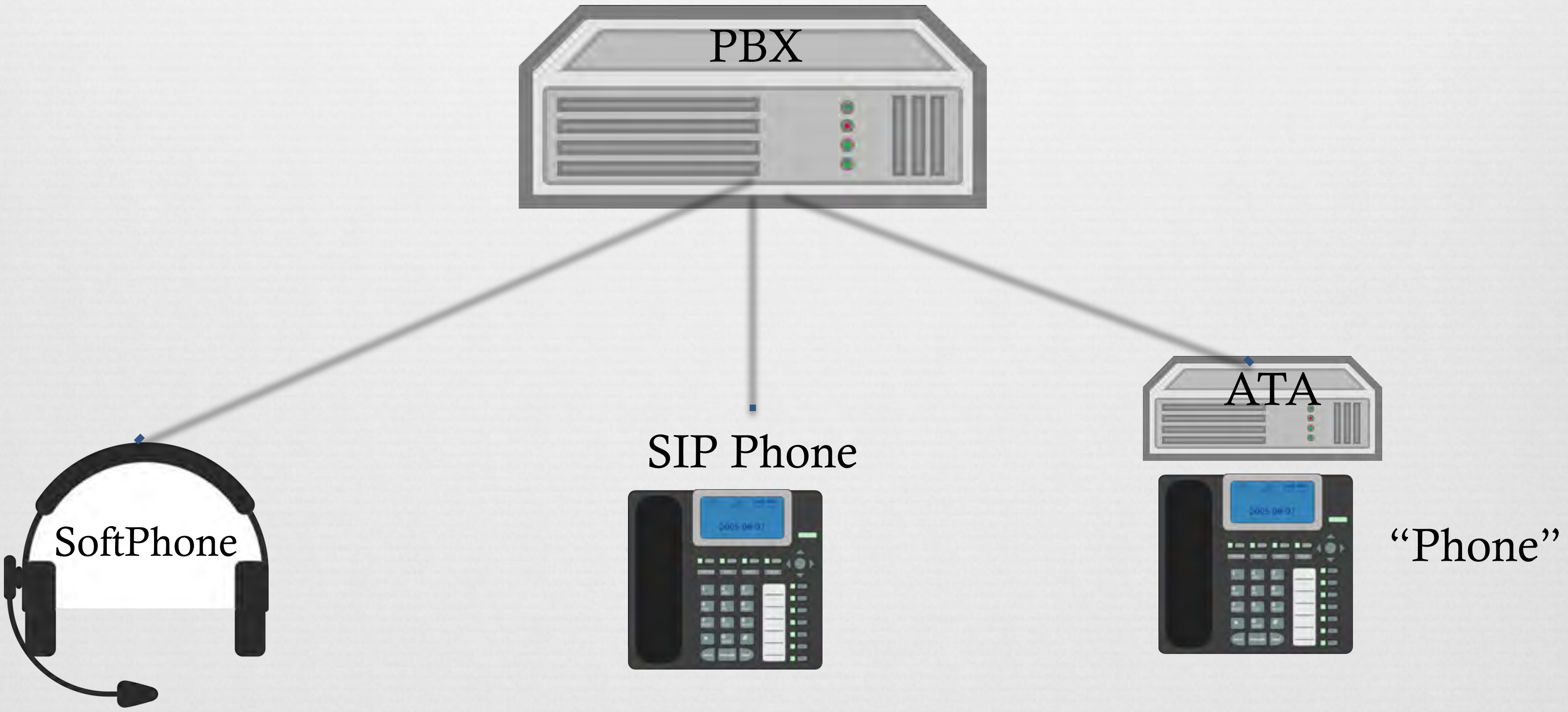
- Hangouts
- FB Messenger
- Signal
- Screen Sharing
- LiveStreaming
- WhatsApp
- SnapChat
- Kik
- etc etc.

PBX’s

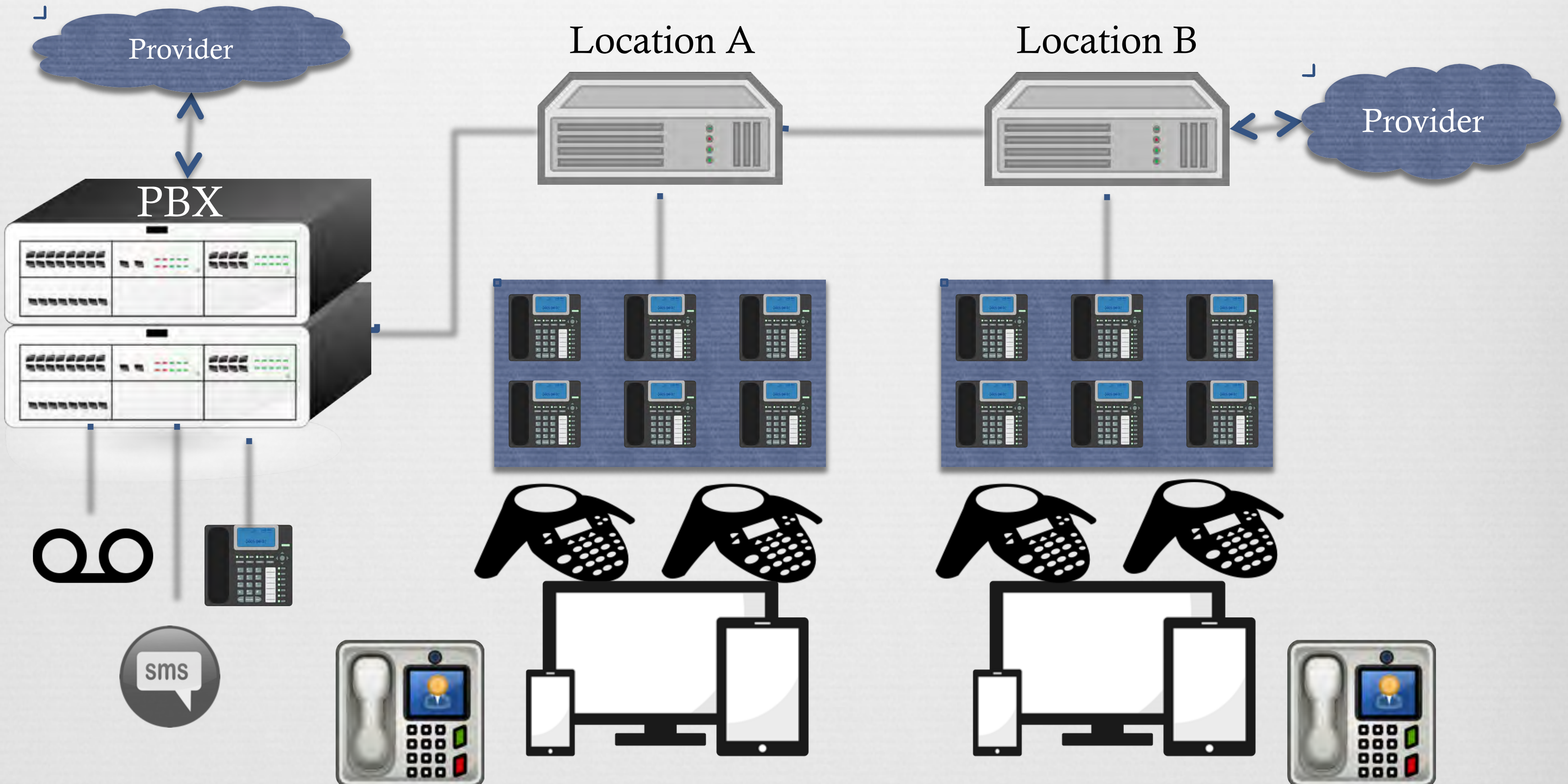
Why do people run PBX’s?

- Reduce Costs
- Cheap Calling
- "Apps”
  - Voicemail
  - IVR’s
  - Conferencing
  - Directories

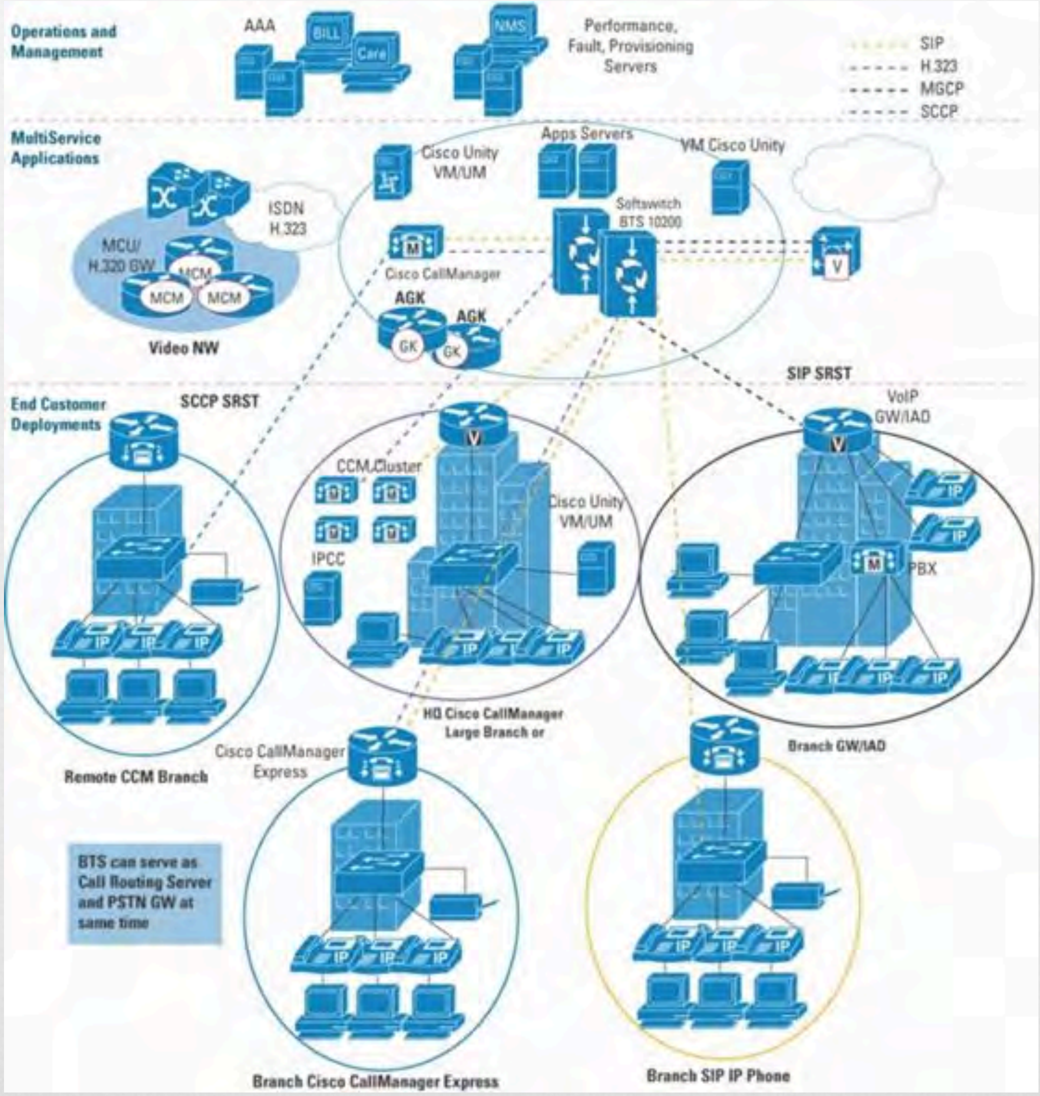
Basic Deployment



Common Deployment



Large



Operations and Management
AAA
BILL
Care
NMS
Performance, Fault, Provisioning Servers
SIP
H.323
MGCP
SCCP
MultiService Applications
Cisco Unity VM/UM
Apps Servers
VM Cisco Unity
ISDN H.323
MCU/ H.320 GW
MCM
Softswitch BTS 10200
Cisco CallManager
AGK
GK
Video NW
SIP SRST
End Customer Deployments
SCCP SRST
VoIP GW/IAD
CCM Cluster
Cisco Unity VM/UM
IPCC
PBX
IP
HQ Cisco CallManager Large Branch or
Cisco CallManager Express
Remote CCM Branch
Branch GW/IAD
BTS can serve as Call Routing Server and PSTN GW at same time
Branch Cisco CallManager Express
Branch SIP IP Phone

More Tech

- Call Monitoring
- Voicemail Transcribing
- Call center / Queue
- Ring Groups
- Call Backs
- Portals
- Reporting and Analytics
- Translations
- Voice Biometrics
- 2FA
- Mobile
  - Forwarding
  - BYOD
  - Apps
- Softphones / Skype

DTMF

Dual Tone
Multi
Frequency

Can be easily generated

	1209 Hz	1336 Hz	1477 Hz	1633 Hz
697 Hz	1	2	3	A
770 Hz	4	5	6	B
852 Hz	7	8	9	C
941 Hz	*	0	#	D

http://www.genave.com/dtmf.htm
https://en.wikipedia.org/wiki/Dual-tone_multi-frequency_signaling

Common Protocols

- SIP
- RTP
- XMPP
- IAX

Codecs

- G.711 – ITU-T
  - PCM
  - Alaw
  - Ulaw
- G.711.0
- G.711.1
- g.722
- GSM

```
*CLI> core show codecs
Disclaimer: this command is for informational purposes only.
        It does not indicate anything about your configuration.
      ID  TYPE      NAME DESCRIPTION
------------------------------------------------------------------------------------
  100001 audio      g723 (G.723.1)
  100002 audio       gsm (GSM)
  100003 audio      ulaw (G.711 u-law)
  100004 audio      alaw (G.711 A-law)
  100011 audio      g726 (G.726 RFC3551)
  100006 audio     adpcm (ADPCM)
  100019 audio      slin (16 bit Signed Linear PCM)
  100007 audio     lpc10 (LPC10)
  100008 audio      g729 (G.729A)
  100009 audio     speex (SpeeX)
  100016 audio   speex16 (SpeeX 16khz)
  100010 audio      ilbc (iLBC)
  100005 audio g726aal2 (G.726 AAL2)
  100012 audio      g722 (G722)
  100021 audio    slin16 (16 bit Signed Linear PCM (16kHz))
  300001 image      jpeg (JPEG image)
  300002 image       png (PNG image)
  200001 video      h261 (H.261 Video)
  200002 video      h263 (H.263 Video)
  200003 video     h263p (H.263+ Video)
  200004 video      h264 (H.264 Video)
  200005 video     mpeg4 (MPEG4 Video)
  400001  text       red (T.140 Realtime Text with redundancy)
  400002  text      t140 (Passthrough T.140 Realtime Text)
  100013 audio    siren7 (ITU G.722.1 (Siren7, licensed from Polycom))
  100014 audio   siren14 (ITU G.722.1 Annex C, (Siren14, licensed from Polycom))
  100017 audio   testlaw (G.711 test-law)
  100015 audio      g719 (ITU G.719)
  100028 audio   speex32 (SpeeX 32khz)
  100020 audio    slin12 (16 bit Signed Linear PCM (12kHz))
  100022 audio    slin24 (16 bit Signed Linear PCM (24kHz))
  100023 audio    slin32 (16 bit Signed Linear PCM (32kHz))
  100024 audio    slin44 (16 bit Signed Linear PCM (44kHz))
  100025 audio    slin48 (16 bit Signed Linear PCM (48kHz))
  100026 audio    slin96 (16 bit Signed Linear PCM (96kHz))
  100027 audio   slin192 (16 bit Signed Linear PCM (192kHz))
```

How?

- Step 1) Figure out what you're testing

Testing

Scope

Blackbox / Whitebox?

Info Gathering

Testing

Info Gathering

- OSINT
  - Grab Phone Numbers from Web / Directories.
  - Look for patterns
- Port Scans
- Shodan
- Use the Web
- Whois has information too!

Externally Testing

Testing Via POTS

- Regular Phone. Sit and press buttons
- Modems and AT commands
- Soft Phones
  - Any of the major ones
  - Ekiga, Twinkle ETC.
- Automatable / Scriptable
  - SipCLI
  - Sip.Js & JSSip
  - MJSip
- Use a PBX

My Testing Setup

- OrangePi 2E
- Decent Specs
- Portable

Software

Armbian
Asterisk

Scripting Utilities

More on this Later!

Types of Issues 2017

A1: Injection
A2: Broken Authentication and Session Management
A3: Cross-Site Scripting (XSS)
A4: Broken Access Control
A5: Security Misconfiguration
A6: Sensitive Data Exposure
A7: Insufficient Attack Protection
A8: Cross-Site Request Forgery (CSRF)
A9: Using Components with Known Vulnerabilities
A10: Under protected APIs

A1: Injection

Injection Points: Web, Voice, SIP, DTMF
Result:

- XSS
- SQL
- Buffer Overflows
- Log Contamination

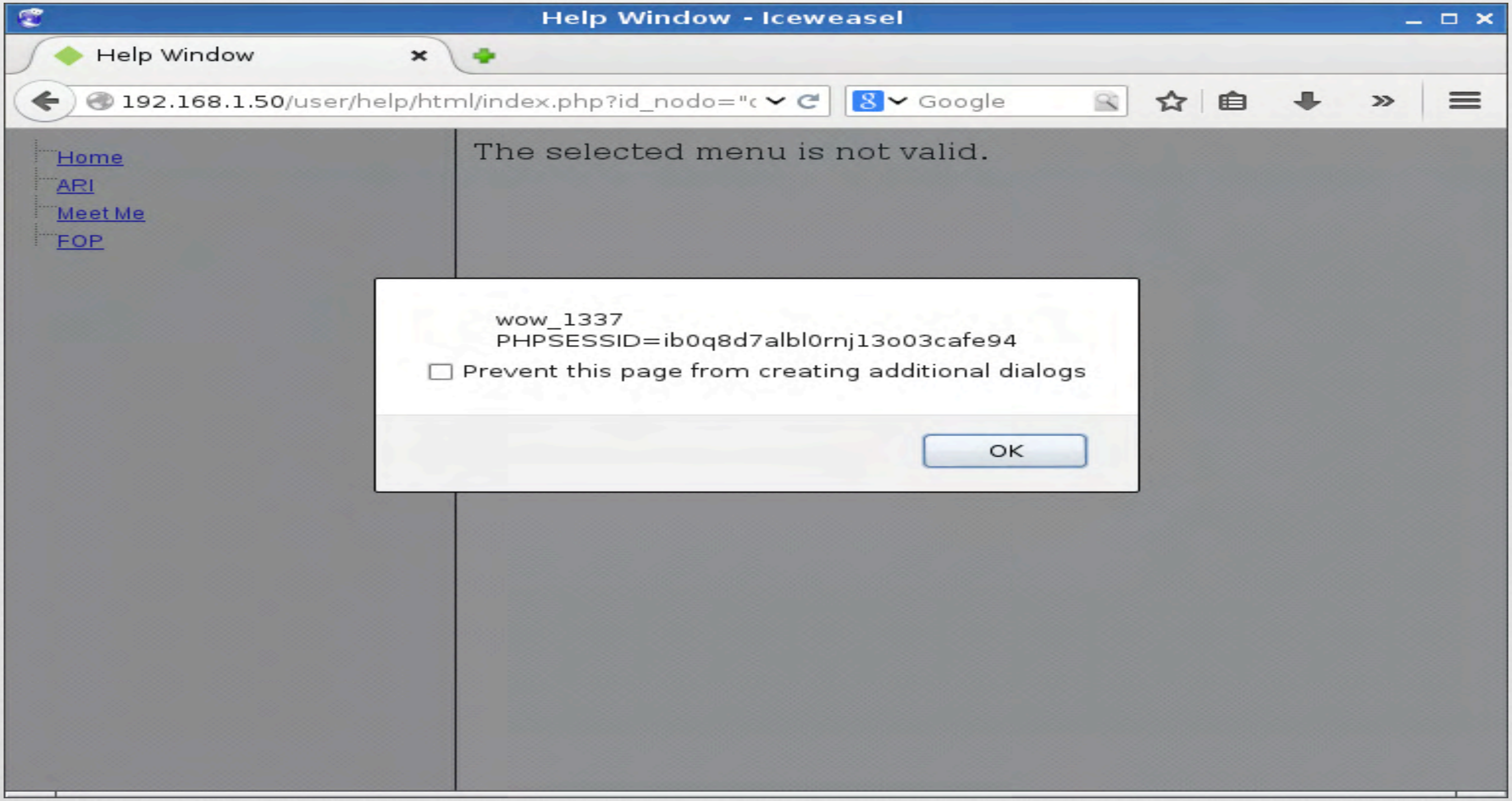
A2: Broken Authentication & Session Management

Mostly Authentication
Lack of SSL/TLS for SIP

https://wiki.asterisk.org/wiki/display/AST/Secure+Calling+Tutorial

A3: Cross-site Scripting

Somewhat covered by injection



A4: Broken Access Control

http://example.com/app/accountInfo?acct=notmyacct

Given that example, this can be translated into a bad configuration.

Either Extensions or AGI Script / App

Related to A5

A5: Security Misconfiguration

- Pretty common
- SIP allowguest – Default = yes
- 4 Digit passwords for SIP Clients
- Conferencing
- Default passwords
- Weak Passwords
- Misconfigured Dial plans & AGI's

A6: Sensitive Data Exposure

- Voicemail
- Conference Calls
- Information not available elsewhere
  - Similar to the User/Password combination enumeration
- Corp Directories
  - Full Names, E-Mails
  - Schedules, out of office

A7: Missing Function Level Access Control

- Caller ID Spoof
- User logs in, tries username / pass, fails tries another.
- Systems like voicemail that allow userid, password separate and prompt for username again is an issue
- Potential with misconfigurations, if put back into another context.
- Reasonable Use

A8: Cross-Site Request Forgery (CSRF)

- Vendors
- Web portals and configuration pages are often vulnerable
- In from a phone sense not directly applicable

A9: Components with Known Vulnerabilities

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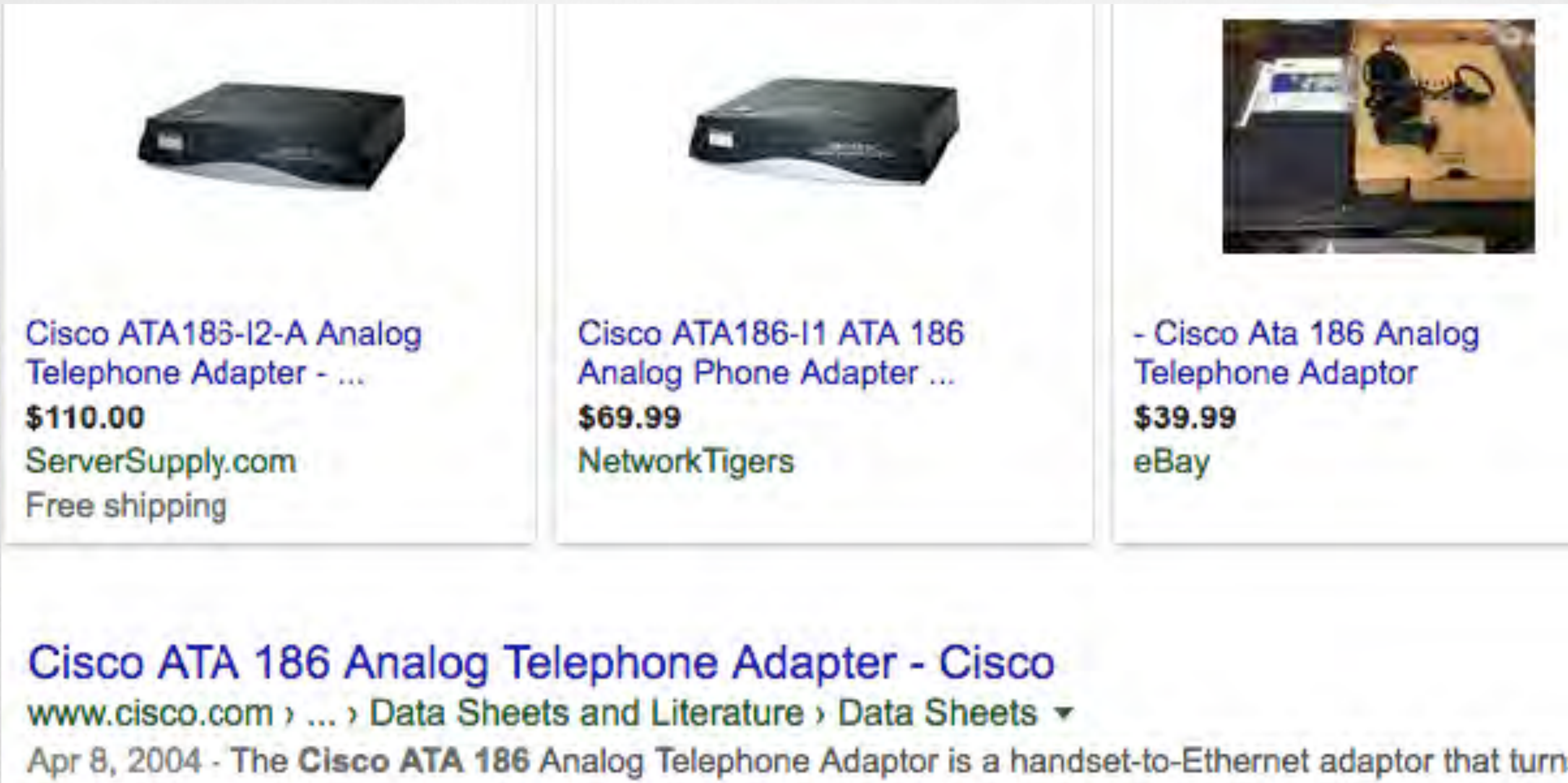
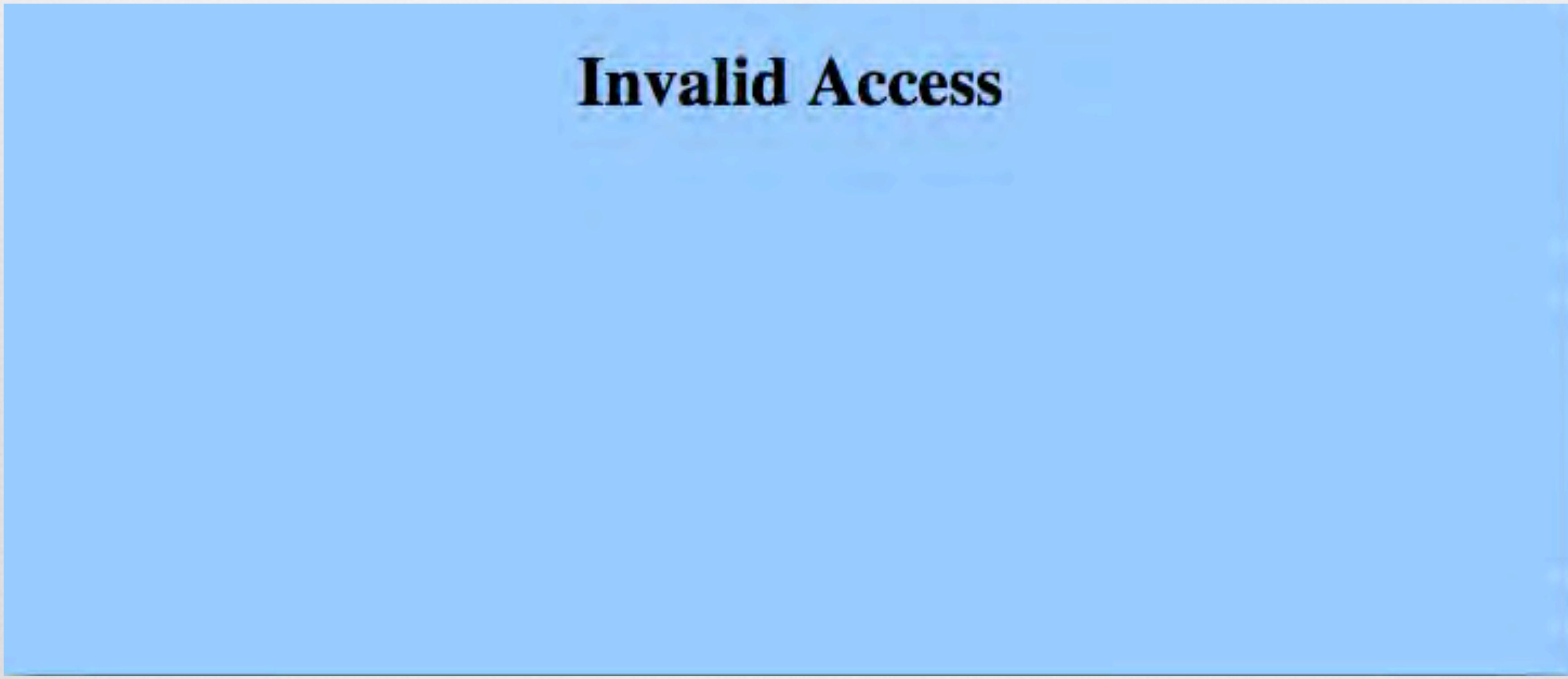


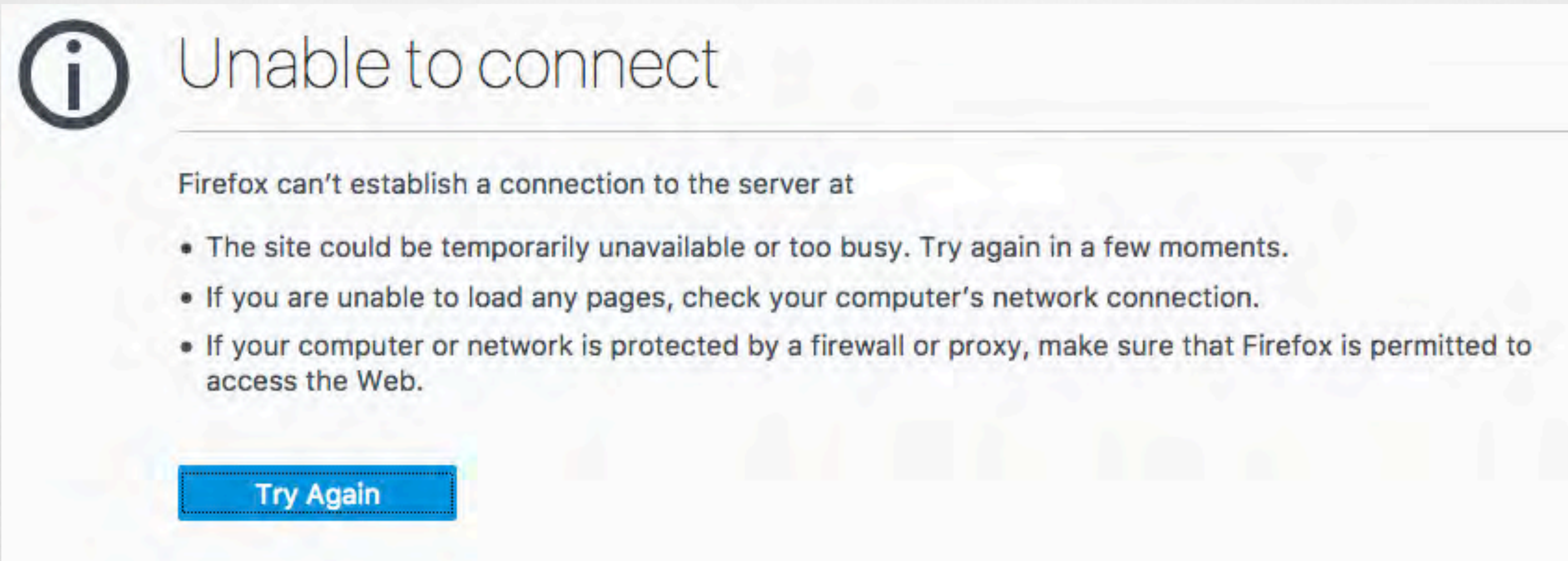
A9: Components with Known Vulnerabilities



A9: Components with Known Vulnerabilities

TftpURL:	[illegible]	CfgInterval:	86400
EncryptKey:	•	EncryptKeyEx:	0000000000000000000C
Dhcp:	0	StaticIP:	
StaticRoute:		StaticNetMask:	
UID0:		PWD0:	
UID1:		PWD1:	
GkOrProxy:		UseLoginID:	0
LoginID0:		LoginID1:	
AltGk:	0	AltGkTimeOut:	0
SIPRegInterval:	120	MaxRedirect:	5
SIPRegOn:	1	NATIP:	0.0.0.0
SIPPort:	5060	MediaPort:	16384
OutBoundProxy:	0	NatServer:	0
NatTimer:	0x00000000	MsgRetryLimits:	0x00000000
SessionTimer:	0x00000000	SessionInterval:	1800
MinSessionInterval:	1800	DisplayName0:	0
DisplayName1:	0	LBRCodec:	0
AudioMode:	0x00140014	RxCodec:	1
TxCodec:	1	NumTxFrames:	2
CallFeatures:	0xffffffff	PaidFeatures:	0xffffffff
CallerIdMethod:	0x00019e60	FeatureTimer:	0x00000000
FeatureTimer2:	[illegible]	Polarity:	[illegible]

A9: Components with Known Vulnerabilities



A9: Components with Known Vulnerabilities

Table 1. End-of-Life Milestones and Dates for the Cisco ATA 186 Analog Telephone Adaptor

Milestone	Definition	Date
End-of-Life Announcement Date	The date the document that announces the end of sale and end of life of a product is distributed to the general public.	March 30, 2010
End-of-Sale Date	The last date to order the product through Cisco point-of-sale mechanisms. The product is no longer for sale after this date.	September 28, 2010
Last Ship Date: HW	The last-possible ship date that can be requested of Cisco and/or its contract manufacturers. Actual ship date is dependent on lead time.	December 27, 2010
End of Routine Failure Analysis Date: HW	The last-possible date a routine failure analysis may be performed to determine the cause of hardware product failure or defect.	September 28, 2011
End of New Service Attachment Date: HW	For equipment and software that is not covered by a service-and-support contract, this is the last date to order a new service-and-support contract or add the equipment and/or software to an existing service-and-support contract.	September 28, 2011
End of Service Contract Renewal Date: HW	The last date to extend or renew a service contract for the product.	December 24, 2014
Last Date of Support: HW	The last date to receive service and support for the product. After this date, all support services for the product are unavailable, and the product becomes obsolete.	September 30, 2015

- http://www.cisco.com/c/en/us/products/unified-communications/ata-180-series-analog-telephone-adaptors/end_of_life_notice_c51-585199.html

A9: Components with Known Vulnerabilities

- How does this apply?

A10 - Underprotected APIs

AGI
ARI
WebRTC
    wss://

OWASP Mapping

A1: Injection	1: Security Misconfiguration
A2: Broken Authentication and Session Management	2: Broken Authentication and Session Management
A3: Cross-site Scripting	3: Injection
A4: Broken Access Control	4: Using Components with Known Vulnerabilities
A5: Security Misconfiguration	5:Broken Access Control
A6: Sensitive Data Exposure	6: Insufficient Access Protection
A7: Insufficient Access Protection	7: Sensitive Data Exposure
A8: Cross-Site Request Forgery (CSRF)	8: XSS
A9: Using Components with Known Vulnerabilities	9: Underprotected API's
A10: Under Protected API's	10: CSRF

Using Asterisk

vagrant up

Soft Phone

Console

AGI

```
contrib-jessie*CLI> core show help core show
```

```
contrib-jessie*CLI> core show help core show help
Usage: core show help [topic]
       When called with a topic as an argument, displays usage
       information on the given command. If called without a
       topic, it provides a list of commands.
```

https://wiki.asterisk.org/wiki/display/AST/Asterisk+13+Command+Reference

Scenario

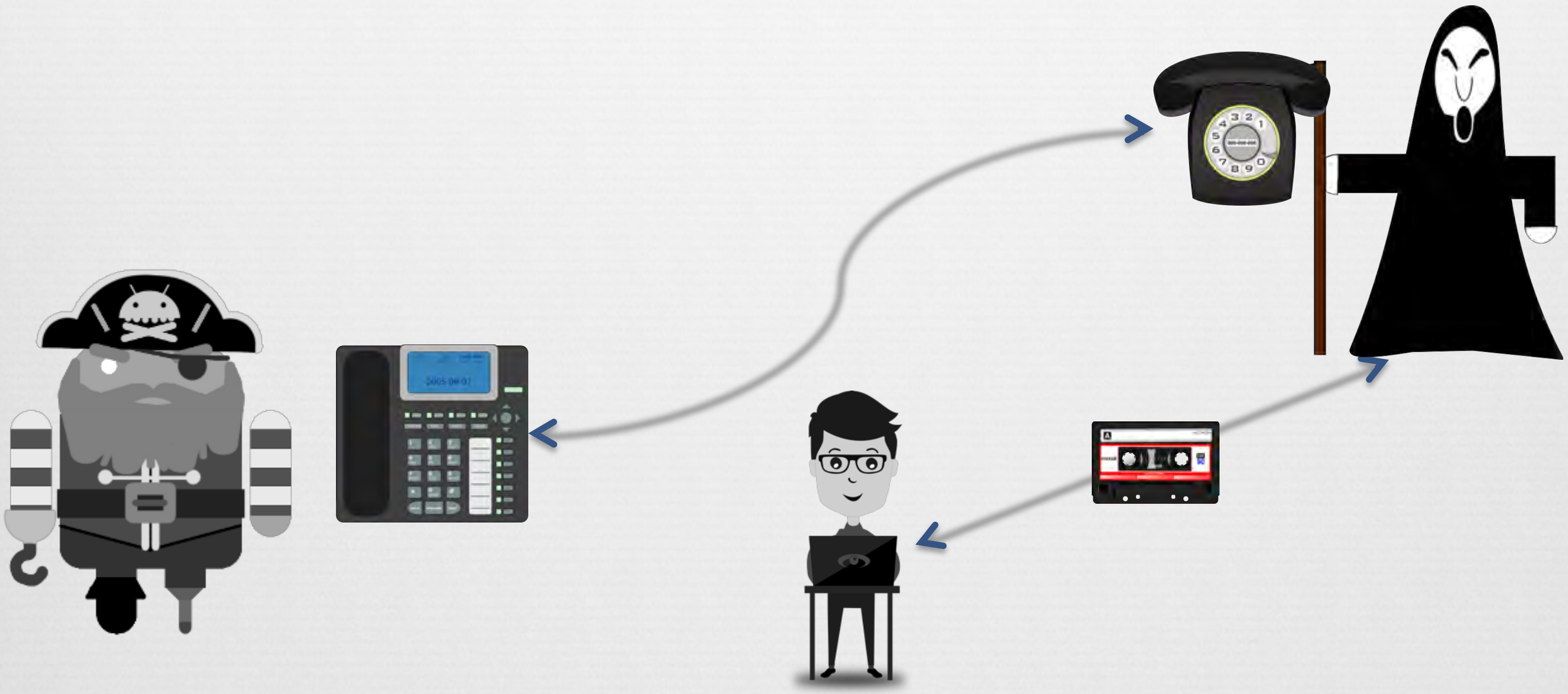
Vectors

Two Vectors

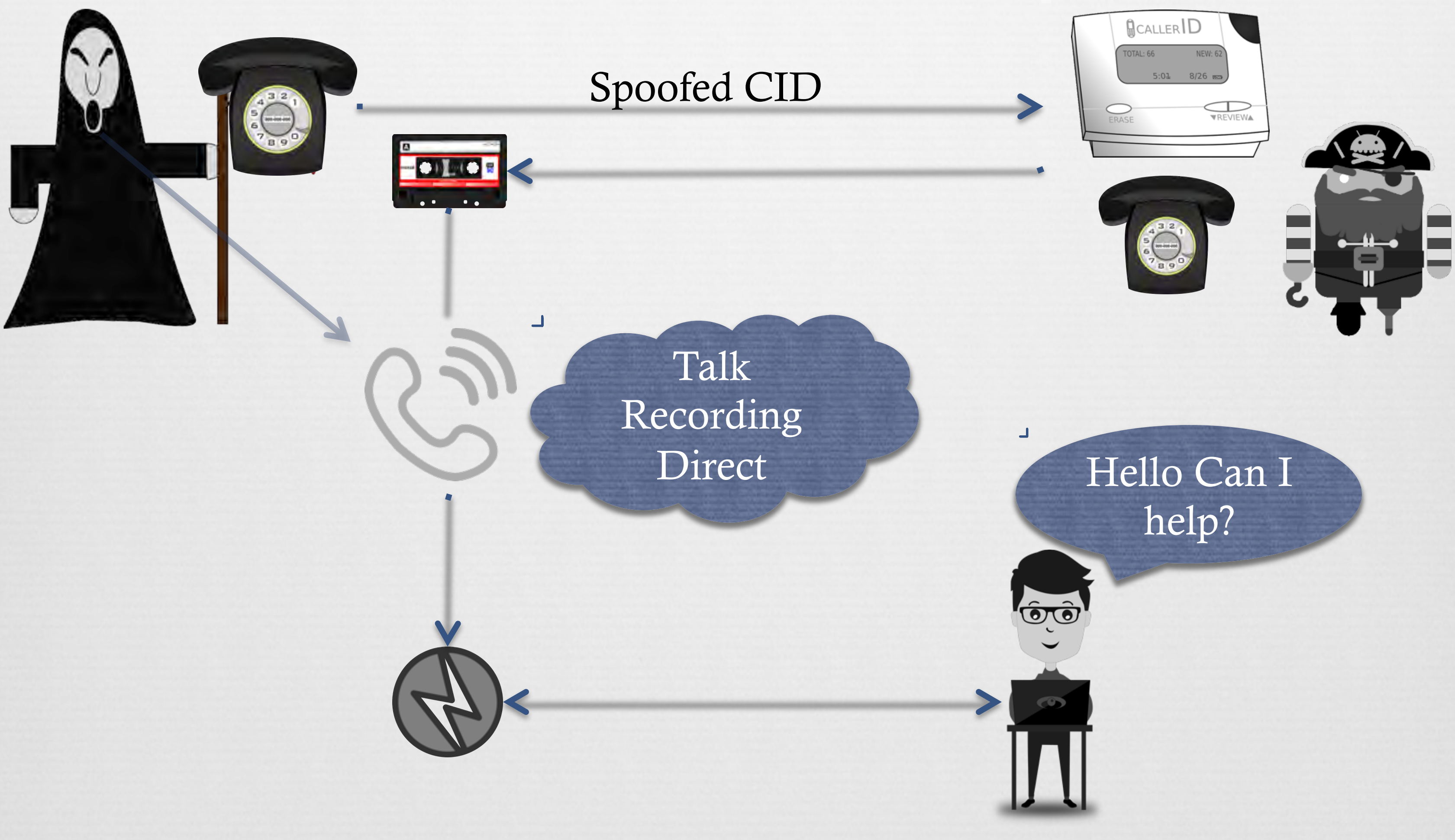
A. Fat Finger Squat

B. Spoofed Target Vish

Vector A - Fat Finger Squat



Vector B – Spoofed Target Vish



Vector A

- Demo Time

Result

Left with a Recording
- What does that contain?

What’s that Sound?

Software

- DTMF Decoding

Software

- Online (dialabc)

Hardware Decoder with

ATA or line out

Tones Found

Tone	Start Offset [ms]	End Offset [ms]	Length [ms]
1	845 ± 15	965 ± 15	120 ± 30
5	1,056 ± 15	1,177 ± 15	120 ± 30
7	1,267 ± 15	1,388 ± 15	120 ± 30
0	1,448 ± 15	1,569 ± 15	120 ± 30
2	1,659 ± 15	1,780 ± 15	120 ± 30
3	1,871 ± 15	1,991 ± 15	120 ± 30
4	2,052 ± 15	2,173 ± 15	120 ± 30
0	2,263 ± 15	2,384 ± 15	120 ± 30
0	2,444 ± 15	2,565 ± 15	120 ± 30
0	2,655 ± 15	2,776 ± 15	120 ± 30
3	2,867 ± 15	2,987 ± 15	120 ± 30

http://dialabc.com/sound/detect/index.html

Phreak Me

PhreakMe (github.com/phreakme)

- Overview
- Last Years Changes
- More Changes to come

Wrap Up